

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	B.Shyam raj	Reg. No.:	192421176
Section / Batch:	2024	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT**Code of Conduct**

I ___B.Shyam raj(192421176)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				

Total Marks (100):	
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Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q85. Brute-Force String Matching

a) How the algorithm works

- The algorithm compares the keyword with every possible position in the document.
- It checks each character one by one until a match is found or all positions are checked.
- If all characters match, the keyword is detected.

b) Time Complexity

- Best Case: $O(n)$ (match found immediately)
- Worst Case: $O(n \times m)$, where n is the text length and m is the pattern length.
- Average Case: $O(n \times m)$

c) Inefficiency for Large Inputs

- Performs many repeated character comparisons.
- Becomes slow for large documents and long keywords.
- More efficient algorithms like KMP or Boyer–Moore are preferred for large datasets.

Q3. Sequential Search

a) How Sequential Search Works

- Start from the first patient record.
- Compare each record with the required patient ID or name.
- Continue until the record is found or the list ends.

b) Performance Analysis

- Best Case: $O(1)$ (record found at the first position)
- Average Case: $O(n)$ (record found in the middle)
- Worst Case: $O(n)$ (record is last or not found)

c) Why Sorting is Not Preferred

- Patient records are frequently added and updated.
- Maintaining a sorted list after every update is time-consuming.
- An unsorted list allows faster insertion and modification, making sequential search a practical choice.